

NIET Greater Noida

Autonomous Institute

AUTOMATE INDIA

Problem Statement: Smart Waste Management Ecosystem

Route Optimization Theme: Smart City & IoT

Category: Hardware + Software Integration

Team Name: ByteFlow

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IDEA TITLE

❖ **Autonomous Waste Segregation & Fleet Optimization System**

❖ **Core Function:** Automatically segregates wet/dry waste at the source and prevents overflowing street bins.

❖ **Features:** Live dashboard tracking capacity and calculating shortest paths for garbage trucks.

❖ **Innovation:** Replaces expensive manual sorting with low-cost, automated edge-hardware.

❖ **Uniqueness:** Bridges physical IoT sensors with Data Structures & Algorithms (Greedy Approach) for dynamic routing.

TECHNICAL APPROACH

AUTOMATIC WASTE SEGREGATION & OVERFLOW MONITORING SYSTEM (2 CONTAINERS – WET & DRY)

- **Entry IR Sensor:** Detects waste insertion at entry funnel.
- **Capacitive Moisture Sensor:** Evaluates analog moisture (0-1024) for Wet vs Dry.
- **MG996R Servo Motor:** Rotates LEFT (Wet Bin) or RIGHT (Dry Bin).
- **Dual HC-SR04 Ultrasonic Sensors:** Measures Wet & Dry Container Fill Levels.
- **Overflow Alerts:** Red LED + Buzzer (Wet $\geq 85\%$) and Yellow LED + Buzzer (Dry $\geq 85\%$).
- **ESP32 Microcontroller:** Transmits telemetry to web dashboard.

FEASIBILITY AND VIABILITY

- **Feasibility:** Built using highly affordable components (Under ■1,500 per unit).
- **Scalability:** Modular design allows easy scaling across residential and commercial sectors.
- **Challenges:** Sensor degradation over time; Network dropouts in dense urban pockets.
- **Solutions:** Future upgrade to ESP32-CAM (AI Vision) for contactless classification.
- **Local Fail-safes:** LEDs and buzzers operate entirely offline if network drops.

IMPACT AND BENEFITS

- **Target Audience:** Municipal Corporations, Sanitation Workers, and Urban Planners.
- **Fuel Savings:** 30% reduction in municipal fleet fuel consumption (Diesel savings).
- **Source Segregation:** 100% source-level segregation, improving recycling efficiency.
- **Wider Impact:** Directly aligns with Swachh Bharat Abhiyan (Clean India Mission).
- **Environmental Benefit:** Reduces urban carbon footprint (estimated 2,400 tons CO₂ saved annually).

RESEARCH AND REFERENCES

- **Research Sources:** CPCB Guidelines on Solid Waste Management.
- **Academic Literature:** IEEE papers on IoT Smart Dustbins and Graph Algorithms.
- **Field Study:** Local news reports detailing mixed waste crisis in Greater Noida.
- **Similar Solutions:** Traditional static-route garbage trucks (In-efficient).
- **Market Comparison:** Standard market Smart Bins (Lack auto-segregation & fleet routing integration).